

Statistical Methods in Artificial Intelligence (SMAI)

End Semester Examination

Total Time: 3 hours Total Marks: 180

Part I: Objective Questions [60 min; $10 \times 6 = 60$ points]

For Multiple Choice Questions (MCQs), one or more options may be correct (tick/select the correct option(s)). For True/False (T/F), write T or F. For Fill In The Blanks (FITBs), write the most precise answer. Steps/details not required.

1. **IPL Wicket-Prediction Classifier.** A franchise builds a binary classifier that, ball by ball, predicts whether the next delivery will result in a wicket. On a test set of 1,000 balls (of which 15 actually produce wickets), the model flags 20 balls as “wicket-likely” and gets 12 of the actual wickets correct. [2+2+2]
 - (a) The confusion matrix entries are: TP = _____, FP = _____, FN = _____, TN = _____.
 - (b) The *precision* of the wicket class equals _____ and the *recall* equals _____.
 - (c) (MCQ) The franchise’s analyst declares the model “96% accurate”. Which are *valid* criticisms?
 - A. Accuracy is misleading because of severe class imbalance.
 - B. A trivial classifier predicting “no-wicket” always would already exceed 95% accuracy.
 - C. F1-score / AP would be a more honest summary for this skewed problem.
 - D. Confusion matrices can never be used when classes are imbalanced.
2. **Linear Regression: Predicting Strike Rate.** A linear regression prediction $\hat{y}_i = \mathbf{w}^\top \mathbf{x}_i$ is fit to predict batter strike-rate from features such as balls faced and boundary count, by minimising MSE over N samples, data matrix X . [2+2+2]
 - (a) (T/F) The closed-form solution $\mathbf{w}^* = (X^\top X)^{-1} X^\top \mathbf{y}$ exists uniquely even when $X^\top X$ is singular. _____
 - (b) The gradient of the MSE objective ($\nabla J =$) with respect to $\mathbf{w}$ equals $\nabla J =$ _____.
 - (c) (MCQ) If the team adds an ℓ_2 regulariser $\lambda \|\mathbf{w}\|_2^2$ (Ridge), then:
 - A. The closed-form becomes $(X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$.
 - B. Some weights are forced to be *exactly* zero.
 - C. Numerical stability of the inverse improves.
 - D. Ridge generally yields *sparser* solutions than Lasso.

3. **PCA / Eigen-Faces.** A research group has $N = 400$ frontal face images of size 100×100 pixels (mean-subtracted) stacked as rows of X . [2+2+2]
- The covariance matrix $\Sigma = \frac{1}{N}X^\top X$ has size _____ $\times$ _____, and its rank is at most _____.
 - (T/F) The non-zero eigenvalues of $X^\top X$ and $XX^\top$ are identical, so it is computationally cheaper here to first eigen-decompose $XX^\top$ and then map eigenvectors back. _____
 - (MCQ) About PCA:
 - It is an *unsupervised* linear method.
 - Its principal directions maximise variance *and*, equivalently, minimise reconstruction error.
 - It is identical to LDA/Fisher when the class labels happen to be balanced.
 - The covariance matrix is symmetric and positive semi-definite.
4. **Bayesian Decision in a Classroom.** A professor classifies each enrolled student as “will attend tomorrow” (ω_1) or “will skip” (ω_2). Past records suggest $P(\omega_1) = 0.7$, $P(\omega_2) = 0.3$, with class-conditional likelihoods $p(x | \omega_1) = 0.4$ and $p(x | \omega_2) = 0.6$ for the observed feature x . [4+2]
- The posterior using Bayes rule $P(\omega_1 | x)$ is _____.
 - (T/F) The Bayes classifier achieves the *minimum possible* probability of error among all classifiers. _____
5. **Naive Bayes & Logistic Regression.** A spam filter for college email uses Naive Bayes over d binary word-presence features. [2+2+2]
- (T/F) Naive Bayes assumes the features are independent. _____
 - (FITB) For logistic regression with the convention $y \in \{0, 1\}$, the model probability is

$$P(y = 1 | x) = \underline{\hspace{2cm}}$$
 and a frequently-used loss is the _____ (a.k.a. negative log-likelihood) loss.
 - (MCQ) Picking the squared-error loss $\frac{1}{2N} \sum_i (\sigma(\mathbf{w}^\top \mathbf{x}_i) - y_i)^2$ for logistic regression is a poor choice because:
 - The objective is non-convex in $\mathbf{w}$.
 - Gradients vanish in the saturated regions of the sigmoid, slowing learning.
 - The cross-entropy / log-loss is the natural MLE for a binary classification model.
 - Squared error always violates Bayes optimality.
6. **SVMs and Kernels.** [3+3]
- (MCQ) For the standard hard-margin SVM $\min \frac{1}{2} \|\mathbf{w}\|^2$ subject to $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$:
 - Only the support vectors have non-zero dual variables α_i .

- B. The geometric margin equals $2/\|\mathbf{w}\|$.
- C. The dual objective depends on data only through inner products $\mathbf{x}_i^\top \mathbf{x}_j$, enabling the kernel trick.
- D. Replacing the constraint by the hinge loss $\max(0, 1 - y_i f(\mathbf{x}_i))$ recovers exactly the same primal problem in the soft-margin case.
- (b) (T/F) For a soft-margin SVM with parameter C , increasing C *reduces* the tolerance for margin violations and tends to produce a classifier with a *narrower* margin. _____
7. **From SLP to MLP.** [3+3]
- (a) (MCQ) A Single-Layer Perceptron:
- A. Is guaranteed to converge in a finite number of steps if the data are linearly separable.
- B. Can solve the 2-input XOR problem with no kernel expansion.
- C. Always converges (with appropriate η) when used as a linear regressor under the delta rule, even with noisy / non-separable data.
- D. Has expressive power equivalent to a sigmoid MLP with one hidden layer.
- (b) (FITB) Consider an MLP with 3 inputs, one hidden layer of 2 sigmoid units, and 1 linear output unit. Every hidden and output unit has a bias. The total number of learnable scalar parameters is _____.
8. **Optimisers, BatchNorm and Dropout.** [2+2+2]
- (a) (T/F) Adam combines momentum (a running first moment of the gradient) with an adaptive per-parameter learning rate (a running second moment), and applies bias-correction to both estimates. _____
- (b) (MCQ) The vanishing-gradient problem in deep networks can be mitigated by:
- A. Using ReLU-family activations instead of saturating sigmoids.
- B. Using residual / skip connections.
- C. Batch normalisation between linear and non-linear layers.
- D. Initialising every weight to zero.
- (c) (MCQ) Which of the following describes the primary original motivation proposed for incorporating Batch Normalization into deep network training?
- A. To stabilize training distributions by reducing Internal Covariate Shift.
- B. To drastically decrease the computational floating-point operations (FLOPs) of the network.
- C. To convert arbitrary non-linear feature maps into strictly linear mappings.
- D. To compress the final serialized model size for mobile-edge deployment.
9. **2D Convolution Arithmetic.** A document-structure recogniser takes a binarised page of size $224 \times 224 \times 1$ as input. [2+2+2]
- (a) (FITB) The first conv layer applies 32 filters of size 7×7 , with stride 2 and padding 3. Spatial output size = _____ $\times$ _____, number of output channels = _____.

- (b) (FITB) The number of learnable parameters in this conv layer (including biases) is _____.
- (c) (FITB) Its output is followed by a 2×2 max-pool with stride 2. The resulting feature-map shape is _____ $\times$ _____ $\times$ _____.
10. **Multihead Self-Attention.** Consider a multihead self-attention layer with 8 heads. Input as well as output are 1,000 vectors (N) of 100 dimensions (d). Query, Key and Value projections are 50-dimensional. There are multiple matrices W_Q , W_K , W_V and W_O . [2+2+2]
- (a) How many total matrices are there? _____
- (b) Write the dimensionality of each of the matrices.
- (c) What is the total number of learnable parameters? _____

Part II: Answer three out of four of the following questions [60 min; $3 \times 20 = 60$ points]

- Backpropagation / Gradient Descent on a Single Neuron.** Consider one sigmoid neuron $o = \phi(w_1x_1 + w_2x_2 + b)$ with $\phi(z) = 1/(1 + e^{-z})$ and squared loss $L = \frac{1}{2}(o - y)^2$. Current parameters: $w_1 = 1$, $w_2 = -1$, $b = 0$. Training sample: $\mathbf{x} = [1, 1]^\top$, $y = 1$. Learning rate $\eta = 1$. [3+10+8]
 - Derive $\phi'(z)$ and express it in terms of $\phi(z)$.
 - Derive expressions using the chain rule for $\partial L/\partial w_1$, $\partial L/\partial w_2$, and $\partial L/\partial b$, and evaluate them numerically.
 - Perform one Gradient Descent (GD) update for the training sample provided, starting from the given parameter values, and report the new (w_1, w_2, b) .
- Bayesian Classification in 1D.** We have two classes A and B each with $p(x | x \in A) = p(x | x \in B) = \mathcal{N}(\mu, \sigma^2)$. We also know $\mu_A = 100$ and $\mu_B = 200$, $\sigma_A^2 = \sigma_B^2 = 100$. We know prior probabilities of class A and B as 80% and 20%. [5+10+5]
 - Write expressions for $p(x | x \in A)$ and $p(x | x \in B)$. Evaluate these two numerically for $x = 150$. Should they sum to 1.0?
 - Write Bayes theorem. Classify $x = 149$, $x = 150$ and $x = 151$ into A or B using the Bayes classification rule. If you have computed pairs of probabilities to compare, should they sum up to 1.0?
 - Numerically compute the posterior probability $P(x \in A | x)$ and $P(x \in B | x)$. Should they sum to 1.0?
- Eigen Decomposition and SVD.** Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

[6+8+6]

- Find both eigenvalues of A and the corresponding unit-norm eigenvectors.
 - Given that you now know the eigenvectors and eigenvalues of A , write the SVD $A = UDV^\top$ explicitly as three matrices U , D and $V^\top$.
 - Compute the best rank-1 approximation A_1 (as discussed in class) $\min_{A_1} \|A - A_1\|^2$.
- 1D K-Means.** [4+6+10]
 - Write the K-Means objective $L = J(\{C_k\}, \{\mu_k\})$ being minimised.
 - Let L^* be the optimal loss when the algorithm converges. How will L^* change if we change K ? Plot a potential graph with K on the X-axis and L^* on the Y-axis. Can/how we use this graph to find the “right” K for a dataset?
 - Consider K-Means with a simple one-dimensional dataset ($d = 1$, $N = 6$ samples): $\{1, 2, 3, 8, 9, 10\}$. Assume $K = 2$ with initial means $\mu_1^{(0)} = 1$, $\mu_2^{(0)} = 2$.

Find the subsets/clusters C_k and their means μ_k . Run iterations until convergence. For each iteration, list (i) cluster assignments, (ii) updated means, and (iii) the value of J . Stop when assignments do not change.

Part III: Answer three out of four of the following questions [60 min; $3 \times 20 = 60$ points]

1. **Parallel / Wider Neural Network with 4 Branches.** The input $\mathbf{x} \in \mathbb{R}^d$ is fed in parallel to *four* branches (see Figure 1). Branch $k \in \{1, 2, 3, 4\}$ has a weight vector $\mathbf{w}_k \in \mathbb{R}^d$ and an output projection (learnable parameter) vector $\mathbf{u}_k \in \mathbb{R}^p$. It first computes the scalar $z_k = \mathbf{w}_k^\top \mathbf{x}$ and $\alpha_k = \sigma(z_k) = \sigma(\mathbf{w}_k^\top \mathbf{x})$ and then contributes $\alpha_k \mathbf{u}_k$ to the final output. [6+7+7]

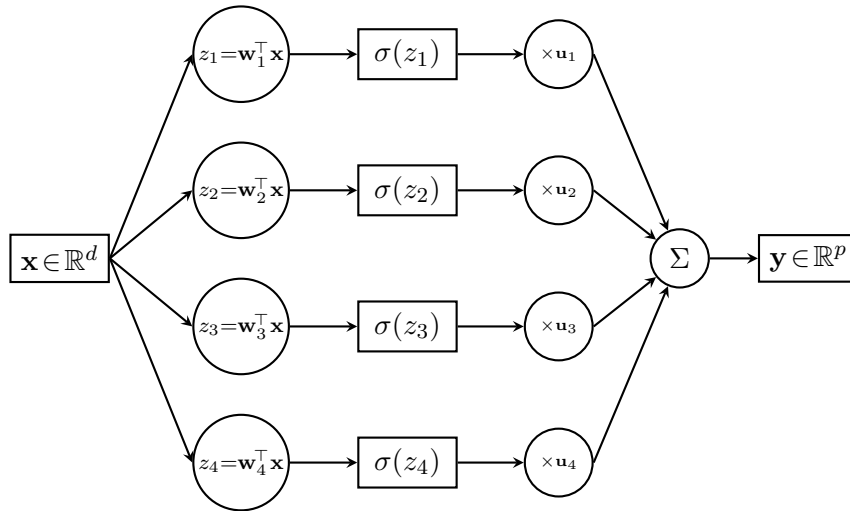


Figure 1: Parallel / Wider Neural Network with 4 Branches

- (a) Write an expression or Python code to compute $\mathbf{y}$ given $\mathbf{x}$.
 - (b) If $d = 1$, compute $\partial y / \partial x$.
 - (c) For general p (and d), compute $\partial y / \partial z_2$.
2. **Kernel Matrix.** Let $\kappa(p, q) = \phi(p)^\top \phi(q)$. For samples $\{x_1, \dots, x_N\}$ define $K_{ij} = \kappa(x_i, x_j)$. [2+8+10]
 - (a) Show that K is symmetric.
 - (b) Show that K is positive semi-definite by proving $\mathbf{v}^\top K \mathbf{v} \geq 0$ for all $\mathbf{v} \in \mathbb{R}^N$.
 - (c) For $p, q \in \mathbb{R}^2$ and $\kappa(p, q) = (p^\top q)^3$, find an explicit feature map $\phi(\cdot)$ such that $\kappa(p, q) = \phi(p)^\top \phi(q)$.
 3. **Backprop Through a Residual Block.** Consider a residual block (see Figure 2). Input is x ; each g_i is a function whose derivative is g'_i . [12+4+4]

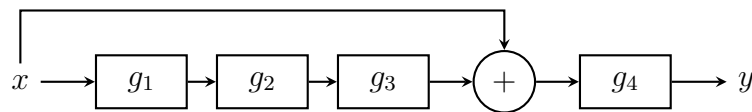


Figure 2: Residual Block

- (a) Write $\partial y / \partial x$ using the chain rule.

- (b) How do residual connections help with the vanishing-gradient problem? Give a single-line answer.
- (c) How does ReLU help with the vanishing-gradient problem?
4. **Triangular Mixture Model (TMM).** You know the Gaussian Mixture Model (GMM) Algorithm. Gaussians have a long tail and also a complex expression (with exponent etc.) Let us assume data is univariate. Assume we simplify the Gaussian as a simple triangle symmetric around the mean, with base:height ratio = 1 : 2 (see Figure 3). The probability is zero outside its base. [6+4+6+4]

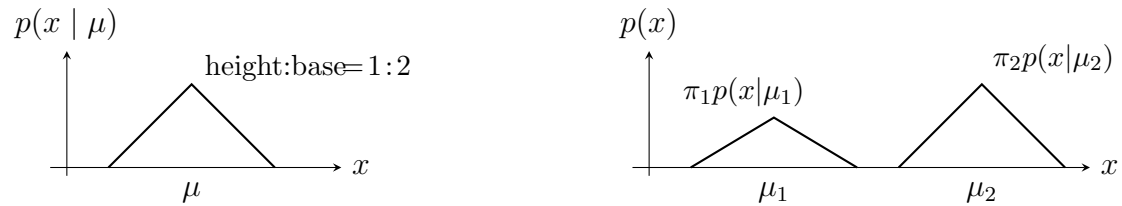


Figure 3: Triangular Mixture Model (TMM): (A) one triangular component $p(x | \mu)$; (B) Triangular Mixture $p(x) = \pi_1 p(x | \mu_1) + \pi_2 p(x | \mu_2)$.

- (a) Write an expression (or Python code) for computing $p(x | \mu)$.
Hint: (i) area under the curve = 1.0; (ii) since the function is symmetric, use of $|x - \mu|$ could simplify.
 Write $p(x | \mu)$ as a piecewise-linear function of $|x - \mu|$, and state the support.
- (b) Recall the GMM algorithm and write its steps with appropriate equations. State the convergence criterion.
- (c) In a similar way, write your own TMM algorithm.
- (d) Why is TMM better/worse compared to GMM? (Write 2–3 advantages and disadvantages for both.)